

Solving Systems by Substitution

Substituting an expression that is ready, isolating a variable first, and reading the two special cases

The pattern to copy

- If one equation already says “ $y = \text{something}$ ” or “ $x = \text{something}$ ”, put that something into the *other* equation in place of the variable. Never back into the same equation it came from — that only tells you $0 = 0$.
 - If neither equation is ready, isolate whichever variable has a coefficient of 1 or -1 ; that keeps the fractions away.
 - Solve for one variable, substitute back to get the other, and write the answer as an ordered pair (x, y) . Check it in *both* original equations.
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1. Solve by substitution: $y = x + 3$ and $2x + y = 12$

Answer: _____

2. Solve by substitution: $x = y - 2$ and $3x + 2y = 9$

Answer: _____

- 3.** Solve by substitution: $y = 2x - 1$ and $4x - y = 7$

Answer: _____

- 4.** Solve by substitution: $x + y = 9$ and $3x - y = 11$

Neither equation is ready, so isolate a variable first.

Answer: _____

- 5.** Solve by substitution: $y = -2x + 9$ and $y = x - 3$

Both equations are already solved for y , so one expression can replace the other directly.

Answer: _____

6. Solve by substitution: $2x + y = 7$ and $3x - 4y = 5$

Answer: _____

7. Consider the system $y = 4x - 2$ and $8x - 2y = 10$.

(a) Substitute and give the solution, or state what the solution set is.

Answer: _____

(b) Explain what the last line of your substitution tells you about this system, and what the two lines look like on a graph.

Answer: _____

8. Solve by substitution: $x - 3y = 1$ and $2x + 5y = 24$

Answer: _____

9. Consider the system $y = 3x - 4$ and $6x - 2y = 8$.

(a) Solve the second equation, $6x - 2y = 8$, for y .

Answer: _____

(b) Now substitute into the second equation. How many solutions does the system have?
Explain how part (a) accounts for what you get.

Answer: _____

10. Solve by substitution: $3x + 2y = 4$ and $x - y = 3$

Answer: _____

11. A booth sold 45 tickets in one evening and took 285 dollars. Adult tickets cost 8 dollars each and child tickets cost 5 dollars each. Let x be the number of adult tickets and y the number of child tickets. Write the two equations and solve the system by substitution to find how many of each kind were sold. Give your answer as the ordered pair (x, y) .

Answer: _____

12. Consider the system $y = \frac{2}{3}x + 1$ and $x + y = 11$.

(a) Solve it by substitution.

Answer: _____

(b) Explain how you could tell, before doing any work, that substitution would be a shorter route here than elimination.

Answer: _____